FILMS DATA BASE USING MONETDB

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COURSE NAME:NO SQL DATABASE

COURSE CODE:CSE-3009

SLOT: L39+L40

1.Create schema

```
sql>CREATE SCHEMA filmdb;
operation successful
```

2.Set schema

```
sql>SET SCHEMA filmdb;
operation successful
```

3.Create directors table

```
sql>CREATE TABLE directors ( director_id INT PRIMARY KEY,director_name
VARCHAR(50),country VARCHAR(30));
operation successful
```

4.**Create films table**

```
sql>CREATE TABLE films (
    film_id INT PRIMARY KEY,
    title VARCHAR(100),
    genre VARCHAR(30),
    release_year INT,
    director_id INT,
    box_office DECIMAL(10,2));
```

operation successful

5.Insert data into directors

5.1 sql>INSERT INTO directors VALUES (1, 'Christopher Nolan', 'UK');
1 affected row
5.2 sql>INSERT INTO directors VALUES
(2, 'Steven Spielberg', 'USA'),
(3, 'Rajkumar Hirani', 'India');
2 affected rows
5.3 sql>INSERT INTO films VALUES (101, 'Inception', 'Sci-Fi', 2010, 1, 829.89);
1 affected row
5.4 sql>INSERT INTO films VALUES (102, 'Interstellar', 'Sci-Fi', 2014, 1, 701.80);
1 affected row
5.5 sql>INSERT INTO films VALUES (103, '3 Idiots', 'Comedy', 2009, 3, 400.50);
1 affected row

6.Display all directors

6.1sql>SELECT * FROM directors;

director_id	director_name	country
1	Christopher Nolan	UK
2	Steven Spielberg	USA
3	Rajkumar Hirani	India

3 tuples

6.2 sql>SELECT * FROM films;

film_id	title	genre	release_year	director_id	box_office
101	Inception	Sci-Fi	2010	1	829.89
102	Interstellar	Sci-Fi	2014	1	701.80
103	3 Idiots	Comedy	2009	3	400.50

3 tuples

6.3 sql>SELECT title, genre FROM films;

```
+--------------+--------+
| title        | genre  |
+==============+========+
| Inception    | Sci-Fi |
| Interstellar | Sci-Fi |
| 3 Idiots     | Comedy |
+--------------+--------+
3 tuples
```

7.WHERE clause

```
sql>SELECT * FROM films WHERE release_year > 2010;
```

```
+---------+--------------+--------+--------------+-------------+--------------+
| film_id | title        | genre  | release_year | director_id | box_office   |
+=========+==============+========+==============+=============+==============+
|     102 | Interstellar | Sci-Fi |         2014 |           1 |       701.80 |
+---------+--------------+--------+--------------+-------------+--------------+
1 tuple
```

8.ORDER BY clause

```
sql>SELECT * FROM films ORDER BY box_office DESC;
```

```
+---------+--------------+--------+--------------+-------------+--------------+
| film_id | title        | genre  | release_year | director_id | box_office   |
+=========+==============+========+==============+=============+==============+
|     101 | Inception    | Sci-Fi |         2010 |           1 |       829.89 |
|     102 | Interstellar | Sci-Fi |         2014 |           1 |       701.80 |
|     103 | 3 Idiots     | Comedy |         2009 |           3 |       400.50 |
+---------+--------------+--------+--------------+-------------+--------------+
3 tuples
```

9.DISTINCT clause

```
sql>SELECT DISTINCT genre FROM films;
```

```
+--------+
| genre  |
+========+
| Sci-Fi |
| Comedy |
+--------+
2 tuples
```

10.LIMIT clause

```
sql>SELECT * FROM films LIMIT 2 OFFSET 1;
```

```
+---------+--------------+--------+--------------+-------------+--------------+
| film_id | title        | genre  | release_year | director_id | box_office   |
+=========+==============+========+==============+=============+==============+
|     102 | Interstellar | Sci-Fi |         2014 |           1 |       701.80 |
|     103 | 3 Idiots     | Comedy |         2009 |           3 |       400.50 |
+---------+--------------+--------+--------------+-------------+--------------+
2 tuples
```

11.LIMIT with OFFSET

sql>SELECT * FROM films LIMIT 2 OFFSET 1;

```
+---------+--------------+--------+--------------+-------------+--------------+
| film_id | title        | genre  | release_year | director_id | box_office   |
+=========+==============+========+==============+=============+==============+
|     102 | Interstellar | Sci-Fi |         2014 |           1 |       701.80 |
|     103 | 3 Idiots     | Comedy |         2009 |           3 |       400.50 |
+---------+--------------+--------+--------------+-------------+--------------+
2 tuples
```

12.COUNT aggregate

sql>SELECT COUNT(*) AS total_films FROM films;

```
+-------------+
| total_films |
+=============+
|           3 |
+-------------+
1 tuple
```

13.SUM aggregate

sql>SELECT SUM(box_office) AS total_collection FROM films;

```
+------------------+
| total_collection |
+==================+
|          1932.19 |
+------------------+
1 tuple
```

14.AVG aggregate

sql>SELECT AVG(box_office) AS average_collection FROM films;

```
+---------------------+
| average_collection  |
+=====================+
|              644.06 |
+---------------------+
1 tuple
```

15.MAX aggregate

```
sql>SELECT MAX(box_office) AS highest_collection FROM films;
```

```
+--------------------+
| highest_collection |
+====================+
|             829.89 |
+--------------------+
1 tuple
```

16.MIN aggregate

```
sql>SELECT MIN(box_office) AS lowest_collection FROM films;
```

```
+-------------------+
| lowest_collection |
+===================+
|            400.50 |
+-------------------+
1 tuple
```

17.GROUP BY clause

```
sql>SELECT genre, COUNT(*) AS total_films FROM films GROUP BY
genre;
```

```
+--------+-------------+
| genre  | total_films |
+========+=============+
| Sci-Fi |           2 |
| Comedy |           1 |
+--------+-------------+
2 tuples
```

18.HAVING clause

```
sql>SELECT genre, COUNT(*) AS total_films FROM films
GROUP BY genre HAVING COUNT(*) > 1;
```

```
+--------+-------------+
| genre  | total_films |
+========+=============+
| Sci-Fi |           2 |
+--------+-------------+
1 tuple
```

19.INNER JOIN

```
sql>SELECT f.title, d.director_nameFROM films f
INNER JOIN directors d ON f.director_id = d.director_id;
```

```
+--------------+-------------------+
| title        | director_name     |
+==============+===================+
| Inception    | Christopher Nolan |
| Interstellar | Christopher Nolan |
| 3 Idiots     | Rajkumar Hirani   |
+--------------+-------------------+
3 tuples
```

20.LEFT JOIN

```
sql>SELECT f.title, d.director_name FROM films f LEFT JOIN
directors d ON f.director_id = d.director_id;
```

```
+--------------+-------------------+
| title        | director_name     |
+==============+===================+
| Inception    | Christopher Nolan |
| Interstellar | Christopher Nolan |
| 3 Idiots     | Rajkumar Hirani   |
+--------------+-------------------+
3 tuples
```

21.CROSS JOIN

```
sql>SELECT f.title, d.director_name FROM films f
CROSS JOIN directors d;
```

22.UPPER function

```
sql>SELECT UPPER(title) FROM films;
```

```
+--------------+-------------------+
| title        | director_name     |
+==============+===================+
+--------------+ Christopher Nolan |
| %1           | Steven Spielberg  |
+==============+ Rajkumar Hirani   |
| INCEPTION    | Christopher Nolan |
| INTERSTELLAR | Steven Spielberg  |
| 3 IDIOTS     | Rajkumar Hirani   |
+--------------+ Christopher Nolan |
3 tuples         Steven Spielberg  |
| 3 Idiots     | Rajkumar Hirani   |
+--------------+-------------------+
9 tuples
```

23.LOWER function

sql>SELECT LOWER(title) FROM films;

```
+--------------+
| %1           |
+==============+
| inception    |
| interstellar |
| 3 idiots     |
+--------------+
3 tuples
```

24.LENGTH function

sql>SELECT LENGTH(title) FROM films;

```
+------+
| %1   |
+======+
|    9 |
|   12 |
|    8 |
+------+
3 tuples
```

25.SUBSTRING function

sql>SELECT SUBSTRING(title, 1, 5) FROM films;

```
+-------+
| %1    |
+=======+
| Incep |
| Inter |
| 3 Idi |
+-------+
3 tuples
```

26.CONCAT function

```
sql>SELECT CONCAT(title, ' - ', genre) FROM films;
```

```
+------------------------+
| concat                 |
+========================+
| Inception - Sci-Fi     |
| Interstellar - Sci-Fi  |
| 3 Idiots - Comedy      |
+------------------------+
3 tuples
```

27.ABS function

```
sql>SELECT ABS(-25);
```

```
+-----+
| abs |
+=====+
|  25 |
+-----+
1 tuple
```

28.ROUND function

```
sql>SELECT ROUND(box_office, 0) FROM films;
```

```
+---------+
| %1      |
+=========+
|  830.00 |
|  702.00 |
|  401.00 |
+---------+
3 tuples
```

29.CEIL function

```
sql>SELECT CEIL(45.3);
```

```
+-------------------------+
| %2                      |
+=========================+
|                      46 |
+-------------------------+
1 tuple
```

30.FLOOR function

```
sql>SELECT FLOOR(45.9);
```

```
+-------------------------+
| %2                      |
+=========================+
|                      45 |
+-------------------------+
1 tuple
```

31.MOD function

```
sql>SELECT MOD(10, 3);
```

```
+------+
| %2   |
+======+
|    1 |
+------+
1 tuple
```

32.POWER function

```
sql>SELECT POWER(2, 3);
```

```
+-------------------------+
| %2                      |
+=========================+
|                       8 |
+-------------------------+
1 tuple
```

33.SQRT function

```
sql>SELECT SQRT(81);
```

```
+-------------------------+
| %2                      |
+=========================+
|                       9 |
+-------------------------+
1 tuple
```

34.CURRENT_DATE

sql>SELECT CURRENT_DATE;

```
+------------+
| %2         |
+============+
| 2026-03-31 |
+------------+
1 tuple
```

35.CURRENT_TIME

sql>SELECT CURRENT_TIME;

```
+-----------------------+
| %2                    |
+=======================+
| 22:07:34.977153+05:30 |
+-----------------------+
1 tuple
```

36.CURRENT_TIMESTAMP

sql>SELECT CURRENT_TIMESTAMP;

```
+----------------------------------+
| %2                               |
+==================================+
| 2026-03-31 22:07:42.343265+05:30 |
+----------------------------------+
1 tuple
```

37.UPDATE command

sql>UPDATE films SET box_office = 850.00 WHERE film_id = 101;

1 affected row

37.1 sql>SELECT * FROM films WHERE film_id = 101;

```
+---------+-----------+--------+--------------+-------------+--------------+
| film_id | title     | genre  | release_year | director_id | box_office   |
+=========+===========+========+==============+=============+==============+
|     101 | Inception | Sci-Fi |         2010 |           1 |       850.00 |
+---------+-----------+--------+--------------+-------------+--------------+
1 tuple
```

38.DELETE command

sql>>DELETE FROM films WHERE film_id = 103;
1 affected row

38.1 sql>SELECT * FROM films;

film_id	title	genre	release_year	director_id	box_office
101	Inception	Sci-Fi	2010	1	850.00
102	Interstellar	Sci-Fi	2014	1	701.80

2 tuples

39.Create VIEW

sql>CREATE VIEW film_report AS SELECT f.title, d.director_name FROM films fJOIN directors d ON f.director_id = d.director_id;
operation successful

40.Display VIEW

sql>SELECT * FROM film_report;

title	director_name
Inception	Christopher Nolan
Interstellar	Christopher Nolan

2 tuples

41.Create INDEX

sql>CREATE INDEX idx_film_title ON films(title);
operation successful

42.Drop INDEX

sql>DROP INDEX idx_film_title;
operation successful

43.EXIT MONETDB

sql>exit